

Q.23: $Ax = B; \quad x + ky = 1$
 $kx + y = 1;$

A (coeff matrix) $= \begin{bmatrix} 1 & k \\ k & 1 \end{bmatrix}$

Augmented matrix $[A|B]$

$$= \begin{bmatrix} 1 & k & 1 \\ k & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1$$

$$\Rightarrow \begin{bmatrix} 1 & k & 1 \\ k+1 & k+1 & 0 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - C_1$$

$$= \begin{bmatrix} 1 & k-1 & 1 \\ k+1 & 0 & 0 \end{bmatrix}$$

I. Unique solution;

$$\text{Rank}(A) = \text{rank}(A|B) = 2$$

$$\det(A) \neq 0;$$

$$1 - k^2 \neq 0 : k \neq 1, -1$$

$$\text{then : rank}(A) = 2$$

$$\text{rank}(A|B) = 2$$

$$k \in \mathbb{R} - \{1, -1\}$$

When $k = 1$

$$A|B = \begin{bmatrix} 1 & 0 & 1 \\ k+1 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix} \rightarrow 0x + 0y = 1$$

$\Rightarrow$ no solution.

Case 3:

$$k = -1$$

$$[A|B] = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank} = 1;$$

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \xrightarrow{R_2 = R_2 + R_1} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$\text{Rank} = 1$$

$$\text{rank}(A) = \text{rank}(A|B) = 1 < n(2)$$

$\Rightarrow$ ∞ solutions.